

Chapter 2: Limits (Continued)

Infinitesimals · *Limit Laws* · *Order of Infinitesimals*

Advanced Mathematics
(Tongji University, 8th Ed.)

Section 4 / 5 / 7

Outline

- 1 Section 4: Infinitesimals and Infinities
- 2 Section 5: Limit Laws (Operational Rules)
- 3 Section 7: Comparison of Infinitesimals

Section 4

Infinitesimals and Infinities

I. infinitesimal

II. infinity

III. Infinitesimals and Infinities between's r



I. Infinitesimal — Definition

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定义 1 (Infinitesimal)

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- $\lim_{x \rightarrow \infty} \frac{1}{x} = 0$, so $\frac{1}{x}$ is an infinitesimal as $x \rightarrow \infty$.
- $\lim_{x \rightarrow -\infty} \frac{1}{\sqrt{1-x}} = 0$, so $\frac{1}{\sqrt{1-x}}$ is an infinitesimal as $x \rightarrow -\infty$.

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Clearly this can hold only when $c = 0$.

Theorem 1: Relation Between Infinitesimal and Limit

定理 2 (Infinitesimal–Limit Relation)

$$\lim_{x \rightarrow x_0} f(x) = A \iff f(x) = A + \alpha(x),$$

where $\alpha(x)$ is an infinitesimal as $x \rightarrow x_0$.

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The same argument applies to other limiting processes ($x \rightarrow \infty$, etc.).

II. Infinity — Definition

定义 3 (Infinity)

For any given $M > 0$, if there always exists $X > 0$ such that whenever $|x| > X$ (or $0 < |x - x_0| < \delta$), we have

$$|f(x)| > M,$$

then we say $f(x)$ is an **infinity** as $x \rightarrow \infty$ (or $x \rightarrow x_0$), denoted by $\lim_{x \rightarrow x_0} f(x) = \infty$ (or $\lim_{x \rightarrow \infty} f(x) = \infty$).

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If in the definition we require $f(x) > M$ (or $f(x) < -M$), then we denote $\lim f(x) = +\infty$ (or $-\infty$).

II. Infinity — Notes

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- But $f(\frac{\pi}{2} + n\pi) = 0$

Therefore $x \rightarrow \infty$ does not make $f(x)$ an infinity!

Example: Vertical Asymptote

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Therefore $\lim_{x \rightarrow 1} \frac{1}{x-1} = \infty$. □

Note

If $\lim_{x \rightarrow x_0} f(x) = \infty$, then the line $x = x_0$ is called a **vertical asymptote** of the curve $y = f(x)$.

III. Relation Between Infinitesimals and Infinities

定理 5 (Reciprocal Relation)

In the same limiting process of the independent variable:

- ① *If $f(x)$ is an infinity, then $\frac{1}{f(x)}$ is an infinitesimal.*
- ② *If $f(x)$ is an infinitesimal and $f(x) \neq 0$, then $\frac{1}{f(x)}$ is an infinity.*

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Note

According to this theorem, questions about infinities can always be transformed into questions about infinitesimals for discussion.

Section 4 Summary

Summary

- ① Definitions of **infinitesimals** and **infinities**.
- ② **Theorem 1:** Relation between infinitesimal and function limit:
 $\lim f(x) = A \Leftrightarrow f(x) = A + \alpha$ where α is infinitesimal.
- ③ **Theorem 2:** Reciprocal relation between infinitesimals and infinities.
- ④ If $\lim_{x \rightarrow x_0} f(x) = \infty$, then $x = x_0$ is a **vertical asymptote**.

Section 5

Limit Laws (Operational Laws)

- I. infinitesimal operation 法 then
- II. limit's 四 then operation 法 then
- III. composite Limits of Functions operation



I. Operations on Infinitesimals — Sum Rule

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Take $\delta = \min\{\delta_1, \delta_2\}$, then when $0 < |x - x_0| < \delta$:

$$|\alpha + \beta| \leq |\alpha| + |\beta| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$$

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Hence $\lim_{x \rightarrow x_0} (\alpha + \beta) = 0$, i.e. $\alpha + \beta$ is infinitesimal.

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Counterexample:

$$\lim_{n \rightarrow \infty} n \left(\frac{1}{n^2 + \pi} + \frac{1}{n^2 + 2\pi} + \cdots + \frac{1}{n^2 + n\pi} \right) = 1$$

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(Each term is infinitesimal, but there are infinitely many of them.)

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Hence $\lim_{x \rightarrow x_0} u\alpha = 0$, i.e. $u\alpha$ is infinitesimal. □

推论 8

- ① A **constant** times an infinitesimal is an infinitesimal.
- ② The product of **finitely many** infinitesimals is an infinitesimal.

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Remark

$y = 0$ is a horizontal asymptote of $y = \frac{\sin x}{x}$.

II. Four Arithmetic Operations on Limits — Sum/Difference

定理 10

If $\lim_{x \rightarrow x_0} f(x) = A$ and $\lim_{x \rightarrow x_0} g(x) = B$, then

$$\lim_{x \rightarrow x_0} [f(x) \pm g(x)] = \lim_{x \rightarrow x_0} f(x) \pm \lim_{x \rightarrow x_0} g(x) = A \pm B$$

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Proof: Since $\lim f(x) = A$, $\lim g(x) = B$, we have

$$f(x) = A + \alpha, \quad g(x) = B + \beta$$

where α, β are infinitesimals.

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$$f(x) \pm g(x) = (A + \alpha) \pm (B + \beta) = (A \pm B) + (\alpha \pm \beta)$$

By Theorem 1, $\alpha \pm \beta$ is also infinitesimal, hence the conclusion holds. □

Corollary: Preservation of Inequality

推论 11

If $\lim_{x \rightarrow x_0} f(x) = A$, $\lim_{x \rightarrow x_0} g(x) = B$, and $f(x) \geq g(x)$, then $A \geq B$.

Corollary: Preservation of Inequality

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If $\lim_{x \rightarrow x_0} f(x) = A$, $\lim_{x \rightarrow x_0} g(x) = B$, and $f(x) \geq g(x)$, then $A \geq B$.

Hint: Let $\phi(x) = f(x) - g(x) \geq 0$. By Theorem 3:

$$\lim_{x \rightarrow x_0} \phi(x) = \lim[f(x) - g(x)] = \lim f(x) - \lim g(x) = A - B \geq 0$$

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Hence $A \geq B$.

Note

Theorem 3 can be generalized to the case of **finitely many functions** (addition/subtraction).

Theorem 4: Product Rule

定理 12

If $\lim_{x \rightarrow x_0} f(x) = A$ and $\lim_{x \rightarrow x_0} g(x) = B$, then

$$\lim_{x \rightarrow x_0} [f(x)g(x)] = \lim_{x \rightarrow x_0} f(x) \cdot \lim_{x \rightarrow x_0} g(x) = AB$$

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Hint: Use the limit-infinitesimal relation and Theorem 2.

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推论 13

- ① $\lim [C \cdot f(x)] = C \cdot \lim f(x)$ $(C = \text{constant})$
- ② $\lim [f(x)]^n = [\lim f(x)]^n$ $(n = \text{positive integer})$

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- ② $\lim [f(x)]^n = [\lim f(x)]^n$ $(n = \text{positive integer})$

Example 2: For polynomial $P_n(x) = a_0 + a_1x + \cdots + a_nx^n$:

$$\lim_{x \rightarrow x_0} P_n(x) = P_n(x_0)$$

Theorem 5: Quotient Rule

定理 14

If $\lim_{x \rightarrow x_0} f(x) = A$, $\lim_{x \rightarrow x_0} g(x) = B$, and $B \neq 0$, then

$$\lim_{x \rightarrow x_0} \frac{f(x)}{g(x)} = \frac{\lim_{x \rightarrow x_0} f(x)}{\lim_{x \rightarrow x_0} g(x)} = \frac{A}{B}$$

Theorem 5: Quotient Rule

定理 14

If $\lim_{x \rightarrow x_0} f(x) = A$, $\lim_{x \rightarrow x_0} g(x) = B$, and $B \neq 0$, then

$$\lim_{x \rightarrow x_0} \frac{f(x)}{g(x)} = \frac{\lim_{x \rightarrow x_0} f(x)}{\lim_{x \rightarrow x_0} g(x)} = \frac{A}{B}$$

Proof idea: Write $f(x) = A + \alpha$, $g(x) = B + \beta$ where α, β are infinitesimal. Let

$$\gamma = \frac{f(x)}{g(x)} - \frac{A}{B} = \frac{B\alpha - A\beta}{B(B + \beta)}.$$

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$$\gamma = \frac{f(x)}{g(x)} - \frac{A}{B} = \frac{B\alpha - A\beta}{B(B + \beta)}.$$

Here the numerator is infinitesimal, denominator $\rightarrow B^2 \neq 0$ (bounded away from 0), so γ is infinitesimal. Hence $\lim_{x \rightarrow x_0} \frac{f(x)}{g(x)} = \frac{A}{B}$. $\square$

(See textbook p.44 for details.)

Theorem 6: Limit Laws for Sequences

定理 15

If $\lim_{n \rightarrow \infty} x_n = A$ and $\lim_{n \rightarrow \infty} y_n = B$, then:

① $\lim_{n \rightarrow \infty} (x_n \pm y_n) = A \pm B$

② $\lim_{n \rightarrow \infty} x_n y_n = AB$

③ If $y_n \neq 0$ and $B \neq 0$: $\lim_{n \rightarrow \infty} \frac{x_n}{y_n} = \frac{A}{B}$

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Hint: Since a sequence is a special kind of function, these follow directly from Theorems 3, 4, 5.

Example 3: Limit of a Rational Function

例 16

Let $R(x) = \frac{P(x)}{Q(x)}$ where $P(x), Q(x)$ are polynomials. If $Q(x_0) \neq 0$, prove that

$$\lim_{x \rightarrow x_0} R(x) = R(x_0).$$

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Proof:

$$\lim_{x \rightarrow x_0} R(x) = \frac{\lim_{x \rightarrow x_0} P(x)}{\lim_{x \rightarrow x_0} Q(x)} = \frac{P(x_0)}{Q(x_0)} = R(x_0)$$



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Note

If $Q(x_0) = 0$, we **cannot** directly use the quotient rule!

Example 4: Factor and Cancel

例 17

Find $\lim_{x \rightarrow 3} \frac{x^2 - 4x + 3}{x^2 - 9}$.

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Find $\lim_{x \rightarrow 3} \frac{x^2 - 4x + 3}{x^2 - 9}$.

Solution: Factor numerator and denominator:

$$\lim_{x \rightarrow 3} \frac{(x-3)(x-1)}{(x-3)(x+3)} = \lim_{x \rightarrow 3} \frac{x-1}{x+3} = \frac{2}{6} = \frac{1}{3}$$

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Key Point

$x = 3$ makes the denominator zero, but after canceling the common factor $(x-3)$, the limit exists.

Example 5: Denominator $\rightarrow 0$, Numerator $\neq 0$

例 18

Find $\lim_{x \rightarrow 1} \frac{2x - 3}{x^2 - 5x + 4}$.

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Find $\lim_{x \rightarrow 1} \frac{2x - 3}{x^2 - 5x + 4}$.

Solution: At $x = 1$: denominator $= 1 - 5 + 4 = 0$, but numerator $= 2 - 3 = -1 \neq 0$.

$$\lim_{x \rightarrow 1} \frac{x^2 - 5x + 4}{2x - 3} = \frac{1 - 5 + 4}{2 - 3} = \frac{0}{-1} = 0$$

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$$\lim_{x \rightarrow 1} \frac{x^2 - 5x + 4}{2x - 3} = \frac{1 - 5 + 4}{2 - 3} = \frac{0}{-1} = 0$$

Therefore:

$$\lim_{x \rightarrow 1} \frac{2x - 3}{x^2 - 5x + 4} = \infty$$

Example 6: Limit at Infinity — “Grab the Leading Term”

例 19

Find $\lim_{x \rightarrow \infty} \frac{4x^2 - 3x + 9}{5x^2 + 2x - 1}$.

Example 6: Limit at Infinity — “Grab the Leading Term”

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Find $\lim_{x \rightarrow \infty} \frac{4x^2 - 3x + 9}{5x^2 + 2x - 1}$.

Solution: As $x \rightarrow \infty$, both numerator and denominator $\rightarrow \infty$.
Divide both by the highest power x^2 :

$$\lim_{x \rightarrow \infty} \frac{4 - \frac{3}{x} + \frac{9}{x^2}}{5 + \frac{2}{x} - \frac{1}{x^2}} = \frac{4 + 0 + 0}{5 + 0 - 0} = \boxed{\frac{4}{5}}$$

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Find $\lim_{x \rightarrow \infty} \frac{4x^2 - 3x + 9}{5x^2 + 2x - 1}$.

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“Grab the leading term”

General Result: Rational Functions at Infinity

For $a_0 b_0 \neq 0$, m, n non-negative integers:

$$\lim_{x \rightarrow \infty} \frac{a_0 x^m + a_1 x^{m-1} + \cdots + a_m}{b_0 x^n + b_1 x^{n-1} + \cdots + b_n} = \begin{cases} \frac{a_0}{b_0}, & \text{if } n = m \quad (\text{e.g., Ex. 6}) \\ 0, & \text{if } n > m \quad (\text{e.g., Ex. 5}) \\ \infty, & \text{if } n < m \quad (\text{see Ex. 7 below}) \end{cases}$$

III. Composite Function Limit — Theorem 7

定理 20

Let $\lim_{x \rightarrow x_0} \phi(x) = a$, and suppose $\phi(x) \neq a$ when $0 < |x - x_0| < \delta_1$. Also let $\lim_{u \rightarrow a} f(u) = A$. Then:

$$\lim_{x \rightarrow x_0} f[\phi(x)] = A$$

III. Composite Function Limit — Theorem 7

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Let $\lim_{x \rightarrow x_0} \phi(x) = a$, and suppose $\phi(x) \neq a$ when $0 < |x - x_0| < \delta_1$. Also let $\lim_{u \rightarrow a} f(u) = A$. Then:

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Proof: $\forall \varepsilon > 0, \exists \eta > 0$: when $0 < |u - a| < \eta$: $|f(u) - A| < \varepsilon$

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Take $\delta = \min\{\delta_1, \delta_2\}$, then when $0 < |x - x_0| < \delta$: $0 < |\phi(x) - a| = |u - a| < \eta$
Hence $|f[\phi(x)] - A| = |f(u) - A| < \varepsilon$. $\square$

Theorem 7 — Extension

Extension

If $\lim_{x \rightarrow x_0} \phi(x) = \infty$, then similarly:

$$\lim_{x \rightarrow x_0} f[\phi(x)] = \lim_{u \rightarrow \infty} f(u) = A$$

Theorem 7 — Extension

Extension

If $\lim_{x \rightarrow x_0} \phi(x) = \infty$, then similarly:

$$\lim_{x \rightarrow x_0} f[\phi(x)] = \lim_{u \rightarrow \infty} f(u) = A$$

This theorem allows us to compute limits of composite functions via substitution.

Example 7: Composite Function Limit

例 21

Find $\lim_{x \rightarrow 3} \sqrt{\frac{x-3}{x^2-9}}$.

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Solution: Let $u = \frac{x-3}{x^2-9}$. From Example 3, we know $\lim_{x \rightarrow 3} u = \frac{1}{6}$. By Theorem 7 (composite limit):

$$\lim_{u \rightarrow \frac{1}{6}} \sqrt{u} = \sqrt{\frac{1}{6}} = \frac{\sqrt{6}}{6}$$

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Solution: Let $u = \frac{x-3}{x^2-9}$. From Example 3, we know $\lim_{x \rightarrow 3} u = \frac{1}{6}$. By Theorem 7 (composite limit):

$$\lim_{u \rightarrow \frac{1}{6}} \sqrt{u} = \sqrt{\frac{1}{6}} = \frac{\sqrt{6}}{6}$$

Hence:

$$\boxed{\lim_{x \rightarrow 3} \sqrt{\frac{x-3}{x^2-9}} = \frac{\sqrt{6}}{6}}$$

Example 8: Two Methods

例 22

Find $\lim_{x \rightarrow 1} \frac{x - 1}{\sqrt{x} - 1}$.

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Find $\lim_{x \rightarrow 1} \frac{x-1}{\sqrt{x}-1}$.

Method 1 (substitution): Let $u = \sqrt{x}$, then $\lim_{x \rightarrow 1} u = 1$.

$$\frac{x-1}{\sqrt{x}-1} = \frac{u^2-1}{u-1} = u+1 \implies \lim_{u \rightarrow 1} (u+1) = 2$$

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Method 2 (rationalization):

$$\lim_{x \rightarrow 1} \frac{(x-1)(\sqrt{x}+1)}{(\sqrt{x}-1)(\sqrt{x}+1)} = \lim_{x \rightarrow 1} (\sqrt{x}+1) = 2$$

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Limit Laws (Operational Rules)

- ① **Infinitesimal operations**
(Th1: sum; Th2: bounded $\times$ infinitesimal)
- ② **Four arithmetic rules**
(Th3: $\pm$; Th4: $\times$; Th5: $\div$)
- ③ **Composite function limit**
(Th7: substitution)

Methods for finding limits:

- ① $x \rightarrow x_0$: direct substitution
(if denom. $\neq 0$); factor & cancel common factor
- ② $x \rightarrow \infty$: divide by highest power (“grab the leading term”)
- ③ Composite: substitute inner variable

Section 7

infinitesimal's compare 较



Motivation: Different Rates of Tending to Zero

As $x \rightarrow 0$, each of $3x$, x^2 , $\sin x$ is an infinitesimal, but:

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As $x \rightarrow 0$, each of $3x$, x^2 , $\sin x$ is an infinitesimal, but:

$$\lim_{x \rightarrow 0} \frac{x^2}{3x} = 0, \quad \lim_{x \rightarrow 0} \frac{\sin x}{3x} = \frac{1}{3}, \quad \lim_{x \rightarrow 0} \frac{\sin x}{x^2} = \infty$$

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Infinitesimals tend to zero at **different rates!**

Definition: Comparison of Infinitesimals

Let α, β be infinitesimals in the same limiting process.

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$\lim \frac{\beta}{\alpha} = 1$: β and α are **equivalent** infinitesimals,
denoted $\alpha \sim \beta$ or $\beta \sim \alpha$.

Common Equivalent Infinitesimals (as $x \rightarrow 0$)

Standard Equivalents

$$\sin x \sim x,$$

$$1 - \cos x \sim \frac{1}{2}x^2,$$

$$\tan x \sim x,$$

$$\sqrt[n]{1+x} - 1 \sim \frac{1}{n}x$$

$$\arcsin x \sim x,$$

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Examples:

- $x^3 = o(6x^2)$ as $x \rightarrow 0$ (higher order)

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Examples:

- $x^3 = o(x^2)$ as $x \rightarrow 0$ (higher order)
- $\arcsin x \sim x$ as $x \rightarrow 0$ (equivalent)
- $1 - \cos x \sim \frac{1}{2}x^2$ as $x \rightarrow 0$ (2nd order)

Example 1: Proving Equivalence

例 23

Prove: as $x \rightarrow 0$, $\sqrt[n]{1+x} - 1 \sim \frac{1}{n}x$.

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Use $a^n - b^n = (a - b)(a^{n-1} + a^{n-2}b + \cdots + b^{n-1})$:

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Use $a^n - b^n = (a - b)(a^{n-1} + a^{n-2}b + \dots + b^{n-1})$:

$$= \lim_{x \rightarrow 0} \frac{(\sqrt[n]{1+x})^n - 1}{\frac{1}{n}x[(\sqrt[n]{1+x})^{n-1} + (\sqrt[n]{1+x})^{n-2} + \dots + 1]}$$

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Use $a^n - b^n = (a - b)(a^{n-1} + a^{n-2}b + \dots + b^{n-1})$:

$$\begin{aligned} &= \lim_{x \rightarrow 0} \frac{(\sqrt[n]{1+x})^n - 1}{\frac{1}{n}x[(\sqrt[n]{1+x})^{n-1} + (\sqrt[n]{1+x})^{n-2} + \dots + 1]} \\ &= \frac{n}{1 + 1 + \dots + 1} = \frac{n}{n} = 1 \end{aligned}$$

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Use $a^n - b^n = (a - b)(a^{n-1} + a^{n-2}b + \dots + b^{n-1})$:

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Hence $\sqrt[n]{1+x} - 1 \sim \frac{1}{n}x$ as $x \rightarrow 0$. □

Theorem 1: Characterization of Equivalence

定理 24

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$$\iff \beta - \alpha = o(\alpha), \text{ i.e. } \beta = \alpha + o(\alpha) \quad \square$$

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$$\iff \beta - \alpha = o(\alpha), \text{ i.e. } \beta = \alpha + o(\alpha) \quad \square$$

Example: As $x \rightarrow 0$: $\sin x = x + o(x)$, $\tan x = x + o(x)$.

Theorem 2: Equivalence Substitution Rule

定理 25

Let $\alpha \sim \alpha'$, $\beta \sim \beta'$, and assume $\lim \frac{\beta'}{\alpha'}$ exists. Then:

$$\lim \frac{\beta}{\alpha} = \lim \frac{\beta'}{\alpha'}$$

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Proof:

$$\lim \frac{\beta}{\alpha} = \lim \left(\frac{\beta}{\beta'} \cdot \frac{\beta'}{\alpha'} \cdot \frac{\alpha'}{\alpha} \right) = \lim \frac{\beta}{\beta'} \cdot \lim \frac{\beta'}{\alpha'} \cdot \lim \frac{\alpha'}{\alpha} = \lim \frac{\beta'}{\alpha'}$$

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Proof:

$$\lim \frac{\beta}{\alpha} = \lim \left(\frac{\beta}{\beta'} \cdot \frac{\beta'}{\alpha'} \cdot \frac{\alpha'}{\alpha} \right) = \lim \frac{\beta}{\beta'} \cdot \lim \frac{\beta'}{\alpha'} \cdot \lim \frac{\alpha'}{\alpha} = \lim \frac{\beta'}{\alpha'}$$

□ **Example:** $\lim_{x \rightarrow 0} \frac{\tan 2x}{\sin 5x} = \lim_{x \rightarrow 0} \frac{2x}{5x} = \frac{2}{5}$

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$$\text{e.g., } \lim_{x \rightarrow 0} \frac{\sin x}{x^3 + 3x} = \lim_{x \rightarrow 0} \frac{x}{3x} = \frac{1}{3}$$

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If $\alpha \sim \alpha'$, $\beta \sim \beta'$, but α and β are **not equivalent**, then $\alpha - \beta \sim \alpha' - \beta'$, and

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$$\text{e.g., } \lim_{x \rightarrow 0} \frac{\tan 2x - \sin x}{\sqrt{1+x} - 1} = \lim_{x \rightarrow 0} \frac{2x - x}{\frac{1}{2}x} = 2$$

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But if $\alpha \sim \beta$, this conclusion may **fail!**

Note (3): Product Replacement Rule

If $\alpha \sim \beta$ and $\varphi(x)$ has a finite limit (or is bounded), then:

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Find $\lim_{x \rightarrow 0} \frac{\tan x - \sin x}{x^3}$.

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Find $\lim_{x \rightarrow 0} \frac{\tan x - \sin x}{x^3}$.

Solution: We cannot replace term-by-term in the numerator directly!

$$= \lim_{x \rightarrow 0} \frac{\tan x (1 - \cos x)}{x^3} = \lim_{x \rightarrow 0} \frac{x \cdot \frac{1}{2}x^2}{x^3} = \frac{1}{2}$$

Example 2: Combined Equivalent Substitution

例 27

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Therefore:

$$\lim_{x \rightarrow 0} \frac{(1+x^2)^{\frac{1}{3}} - 1}{\cos x - 1} = \lim_{x \rightarrow 0} \frac{\frac{1}{3}x^2}{-\frac{1}{2}x^2} = -\frac{2}{3}$$

Section 7 Summary

1. Comparison of Infinitesimals

Let α, β be infinitesimals in the same limiting process ($\alpha \neq 0$):

$$\lim \frac{\beta}{\alpha} = \begin{cases} 0, & \beta \text{ is higher order than } \alpha \\ \infty, & \beta \text{ is lower order than } \alpha \\ C (\neq 0), & \beta, \alpha \text{ are same order} \\ 1, & \beta, \alpha \text{ are equivalent} \end{cases}$$

Also: $\lim \frac{\beta}{\alpha^k} = C \neq 0$ means β is k -th order w.r.t. α .

Section 7 Summary (cont.)

2. Common Equivalent Infinitesimals (as $x \rightarrow 0$)

$$\sin x \sim x, \tan x \sim x, \arcsin x \sim x, 1 - \cos x \sim \frac{1}{2}x^2, \sqrt[n]{1+x} - 1 \sim \frac{1}{n}x$$

Section 7 Summary (cont.)

2. Common Equivalent Infinitesimals (as $x \rightarrow 0$)

$$\sin x \sim x, \tan x \sim x, \arcsin x \sim x, 1 - \cos x \sim \frac{1}{2}x^2, \sqrt[n]{1+x} - 1 \sim \frac{1}{n}x$$

3. Key Theorems

- ① **Th1:** $\alpha \sim \beta \iff \beta = \alpha + o(\alpha)$
- ② **Th2 (Substitution):** If $\alpha \sim \alpha', \beta \sim \beta'$, then $\lim \frac{\beta}{\alpha} = \lim \frac{\beta'}{\alpha'}$
- ③ Valid for **multiplication/division**; use caution with sums/differences.